

Control System for a Segway

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I. INTRODUCTION

The goal of this paper is to describe the processes of designing a model and a controller for a small model segway. The Segway consists of two wheels attached on either side of a slender frame. In its uncontrolled state, the Segway is unstable, and cannot retain the desired upright position without falling over. Therefore, sensors present on the device will be simulated and implemented into a controller such that this system can be made controllable. To give this assignment more direction, three goals have been defined in regard to the operation of the segway.

- Free-standing stabilisation such that the segway can remain in a 50cm radius for over 30 seconds
- Robust real-time control such that given some disturbance or uneven surface, balance can be retained
- Implementation of a gamepad such that the segway can be ordered to move and turn in real-time

In relation to these goals, a model, state estimator and controller will be developed.

II. ANALYSIS

For consistency, the axis of the Segway will be defined as Y being positive down in the direction of gravity when upright, and Z and X defined as seen in Fig. 1 with the positive Z direction being the "forward" direction.



Fig. 1: Picture of the Segway used, from [1]

A control system is used to stabilize the segway, which monitors its angle and applies a torque on the wheels when the segway starts to fall over. This allows it to stay balanced. To monitor the angle of the segway relative to the ground,

a state estimator, that utilizes the onboard accelerometer and gyroscope, must be constructed. Based on this estimated angle, a controller that is designed using a model of the system, determines the necessary amount of voltage input to the wheels needed to balance the segway.

A. Model

The fundamental properties of the system can be built upon an augmented model for an inverted pendulum. An inverted pendulum has its centre of mass directly above its pivot point, as result it is unstable and will fall over without additional help. The definitions of positive movement directions and forces are shown in Fig. 2

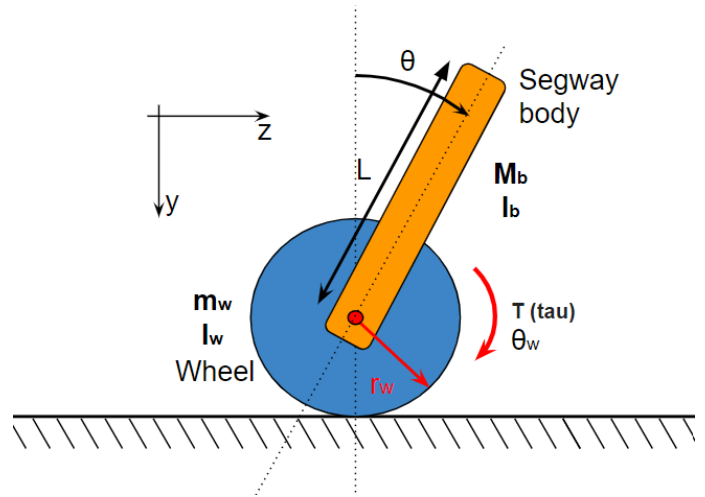


Fig. 2: Basic description of segway behavior from an augmented inverted pendulum

The model can be broken down into three main sections/domains, with each defined by its corresponding energy storage element. The electrical domain with an output of torque feeds the rotational domain, which alongside gravity constitutes the translational domain. By simulating the electrical circuit, the torque on the body from the wheels, and taking into account the masses and inertias of the wheels and segway, a model of the system can be developed. The bond graph of the developed model can be seen in Fig. 3

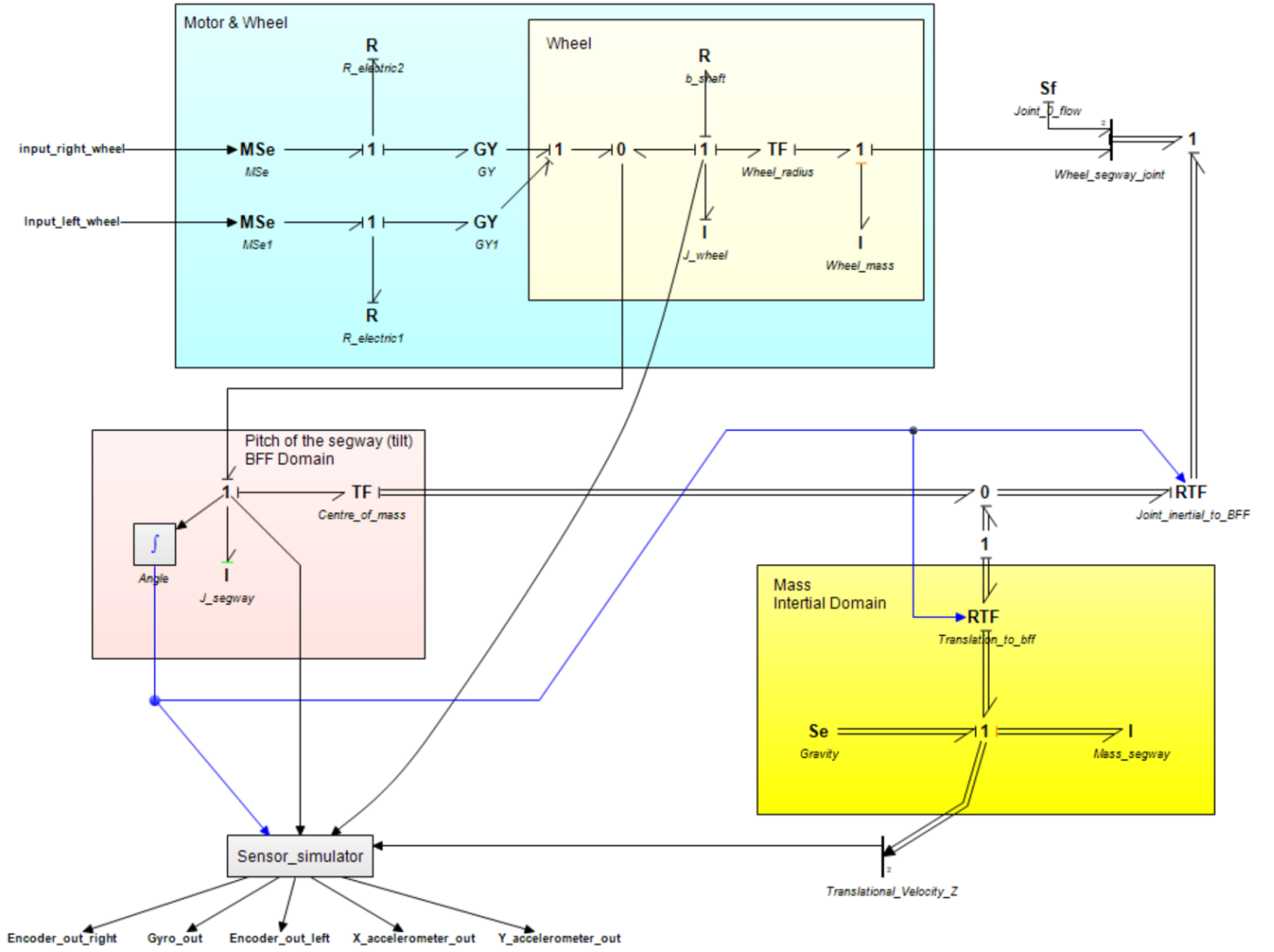


Fig. 3: Bond graph power and signal flow of the model

The model in Fig. 3 above, has the three primary domains separated and labelled respectively. Signal flows, as opposed to power flows, are represented by lines which aren't strictly at 90° angles. Although individual voltage inputs for the motors are present, the model is limited to only planar motion for simplicity (no Y-rotation or X-movement). Traditionally, the model would contain an additional energy storage element in the electrical domain, namely an inductor but was removed due to its small value, and low impact on the simulated data [2], beyond its minor effect as a (high-cutoff) lowpass filter for the voltage inputs. Excluding this allowed for a 3-state controller as will be discussed later. The three domains are interconnected through a Body Fixed Frame (BFF) with a BFF 0-flow joint connecting the translational movement of the wheels to the rotational and translational movement of the segway body.

Additionally to the stated translation, torque from the motor is also directly applied to the body equal to that applied to the wheels, as seen with the 0-junction in the motor section. The torques and angular movements are defined as positive in the clockwise direction throughout the model, resulting in a negative wheel radius used in the TF block connecting the wheel mass to the BFF joint, assuming no-slip meaning $z(t) = r_w \theta_w$. Due to the expected low oscillations and speed of the segway, the air resistance was also neglected in this model for simplicity alongside any flex in the body (rigid body). From this developed model, alongside the parameters determined in the Motor Report [2], the Inertia, mass and centre of mass (COM) of the segway, and wheel radius and wheel mass have to be determined experimentally.

B. State Estimator

The three storage elements, and combined voltage input results in a minimum of one input and three outputs required to fully describe the state of the system, however, the dynamics can be described with the single output of an angle, hence making this a SISO system.

The three desired states will have to be derived from the raw sensor outputs of the segway for full state control. The angular momentum of the wheels and the angular momentum of the segway can be determined from the wheel encoders as done previously in the Motor Report[2], whereas the segway angle will require additional processing. The IMU on the segway outputs gyroscope data which provides high frequency and high precision data, yet suffers from drift when integrated. The IMU also outputs the acceleration of the segway in the Y and Z directions which, assuming no angular velocity, can be used to measure the angle using gravity components. In a sensor fusion setup, the low-frequency offset of the gyro and the high-frequency disturbances of the accelerometers can be accounted for using complementary filtering. A combination of a high-pass filter (HPF) and a low-pass filter (LPF) of equal cut-off for the gyro and accelerometers respectively will result in an evenly filtered output signal.

In the model, this was implemented by means of reading exact flows from one-junctions for the storage elements for each required state. Alongside the modelled angle of the segway, the normalised outputs of each of the used IMU sensors were simulated as shown in Fig. 4 such that a state estimator could also be modelled and any errors could be taken into account for the controller design.

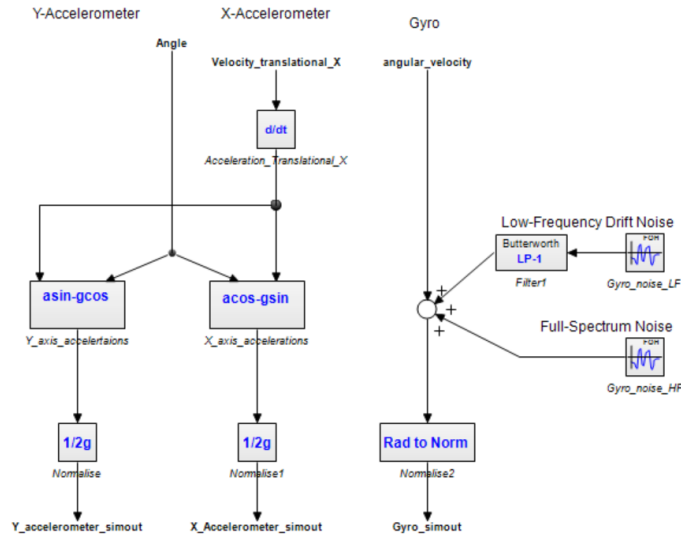


Fig. 4: Simulation of IMU sensor outputs from model states

Each of the simulated accelerometer axes includes what is essentially an RTF block. This block converts the inertial frame translational movement, the force of gravity and the angle of the segway into summed BFF acceleration data as would be read by the real IMU accelerometers on the segway. Using

data from a simple trial of collecting steady-state data from the segway's sensors to determine noise levels, Fig. 5, the noise of the gyroscope was modelled using summed low-frequency and full-spectrum frequency noise blocks, whereas the noise in the accelerometer data was very low so no additional noise was modelled.

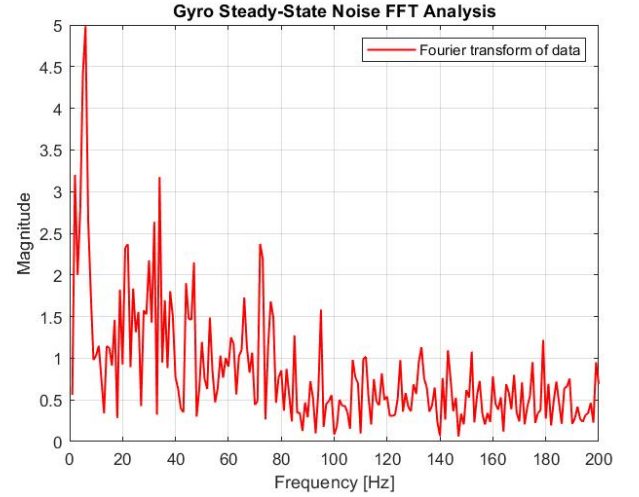


Fig. 5: Fourier transform of static gyroscope data showing significant low frequency (DC) noise and a uniform level of high-frequency noise

Once the filters had been fully simulated, the state estimator could be designed via a complementary filter as seen in Fig. 7 for segway angle, and segway angular momentum. The atan2 block was used to combine the x and y accelerator data into one unified angle. Using the atan2 function also accounted for the issue of the Y accelerometer not outputting unique acceleration data depending on the direction of the tilt as opposed to the X accelerometer, which is an issue that would have persisted if other considered methods of combining the two accelerometers were used. This was the case as instead of the Y accelerometer being used to generate a certain angle independently, it is used in conjunction with the Z acceleration as the vertical position of the segway, shown in Fig.6

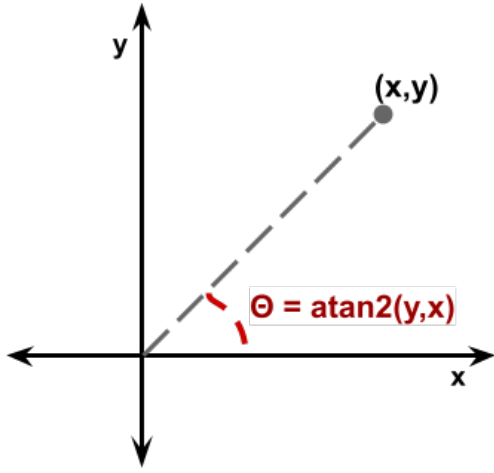


Fig. 6: Graphical representation of the atan2 function used for finding angle from the Z and Y acceleration data

this output angle was then filtered to better approximate the accurate steady-state behaviour of the setup using a LPF. To account for the low-frequency drift of the gyro, after integration, a HPF was applied which then was summed with the filtered accelerometer output to form the final filtered angle output. The cutoff frequency was at this moment chosen as 8Hz based on the collected gyro noise data in Fig. 5, however, will be tuned further with both simulations and by analysing the behaviour of the segway.

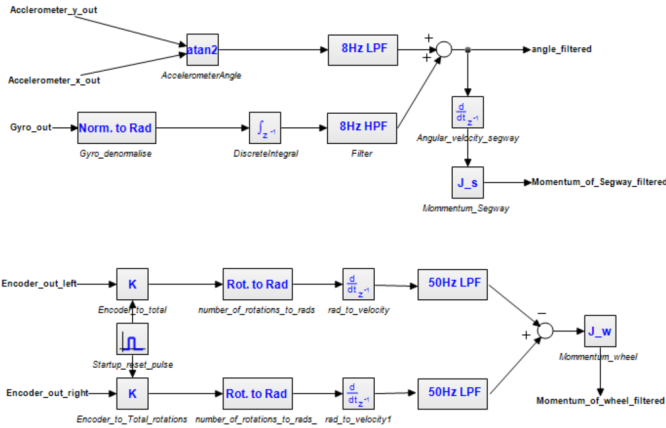


Fig. 7: Block model of the state estimator, with discrete signal inputs and outputs

C. Control theory

The goal of the controller is to keep the segway, an inherently unstable system, in an upright position. A controller $C(s)$ must be able to provide a bounded angle output when a bounded voltage input is applied to the motors. The stability will allow the system to reach a steady state, being the upright position. The model of the segway described above, is a non-linear time invariant open-loop system, meaning that the

system must first be linearized, before applying any type of control.

Linearization of the open-loop model is computed using the "frequency domain toolbox", in 20-sim. The desired input is the voltage applied to the motors, because that is how the segway will be controlled. On the other hand, the output is the angle of the body w.r.t the reference from specified in the model, since this is our desired output. The linearized transfer function can be seen in equation 1.

$$P(s) = \frac{0.05649s^2 + 2.41 \times 10^{-10}s + 4.75 \times 10^{-24}}{s^3 + 0.00325s^2 - 36.8s - 3.092 \times 10^{-07}} \quad (1)$$

Taking a closer look at the transfer function, it can be seen that the system is 3rd order, which corresponds to the intended design of the model. Consider the root locus of the plant in Fig. 8.

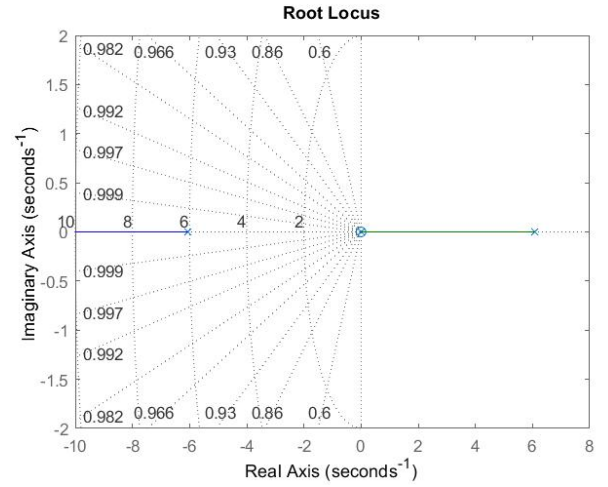


Fig. 8: Root locus of linearized open-loop system, with poles at -6.0678, 6.0645, 0.0 and two zeros -0.4264×10^{-8} and 0.0.

From the root locus, it can be seen that there is one pole with positive real part, that causes the system to be unstable. This must be accounted for during the design of the controller. Furthermore, both zeros present in the transfer function are negative, suggesting that the system is minimum phase and that the root locus method could be applied. However, according to [1], pole cancellation involved in the root locus method is very impractical in real life applications and is therefore not a viable option. Pole placement has also been disregarded for that reason. Other controller options that could be implemented are PID¹ and LQR². First, the Popov-Belevitch-Hautus controllability test [1] is conducted to check whether the system is controllable. It states that if an eigenvalue λ_i becomes a pole, then it is controllable. Consider equation 2:

$$\text{rank}(A - \lambda_i I | B) = n = 3 \quad (2)$$

¹proportional-integral-derivative controller

²Linear-quadratic regulator

As shown in equation 2, the resulting rank is 3 meaning the system is controllable. Since the system is SISO³, PID control is a viable option. However, the drawback of implementing a PID controller for this situation is lack of prioritizing the angle over the other state variables such as momentum of the wheel and momentum of the segway. This is a feature of the LQR controller that will allow for more effective control of the segway, over the alternative PID. LQR minimizes a cost function using weighting factors [3], meaning that it can reduce the voltage input to remain within the 24V limit of the motors. Another benefit is the static gain, meaning the closed-loop system is the same order as the open-loop and the robustness, which achieves an infinite gain margin. The block diagram, in Fig. 9, visualizes the feedback control.

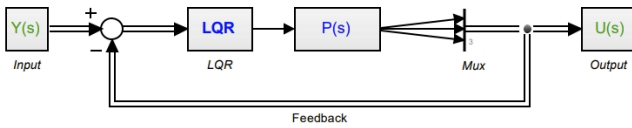


Fig. 9: LQR Block diagram feedback control

The method for designing the LQR controller will be based on simulated and practical performance. This is due to the assumptions made in the model that prevent the simulation from being a perfect representation of practical performance. Therefore, the LQR values must be tuned after analysing the real-life performance of the segway as well.

D. Gamepad

The final component which has to be designed and implemented into the segway system to meet all outlined goals is the wireless gamepad. This gamepad will allow for the segway to be controlled such that in addition to stabilising, it will be able to move around and turn. Due to the simulation being planar this will not be implemented there, and will therefore only be implemented as the last step in the design once a control loop is designed.

There are two main methods forward motion may be implemented, either by applying a certain voltage to both motors or by changing the reference angle. To prevent issues if the controller requires the full $\pm 24V$ range for control, it is preferred to use the reference angle method instead, where the target reference angle for the controller to set the segway as is changed depending on the desired speed and direction. The 20-sim implementation of this can be seen in Fig. 10. Additionally, the reference state of the wheels could be changed instead, but due to potential differences in the motor parameters, and hence wheel states, this was not attempted to avoid any potential unwanted turning.

Due to the states of the two wheels being linked, their reference cannot be used to achieve Y-axis rotation. Since no

measured system state depends on this velocity, the voltage will have to be applied directly to the wheel outputs. Since each motor is pointed in opposite directions, applying the same voltage to both would result in a net torque around the Y-axis and hence a turning motion. If the issue of voltage clipping causing instability during turning appears, a priority system can be implemented such that no turning voltage is applied to the wheels when the full 24V is required for stability. Overall, although this method might encounter issues resulting in not ideally smooth turning around exactly the Y-axis, it would have little impact on the stability of the system during turns.

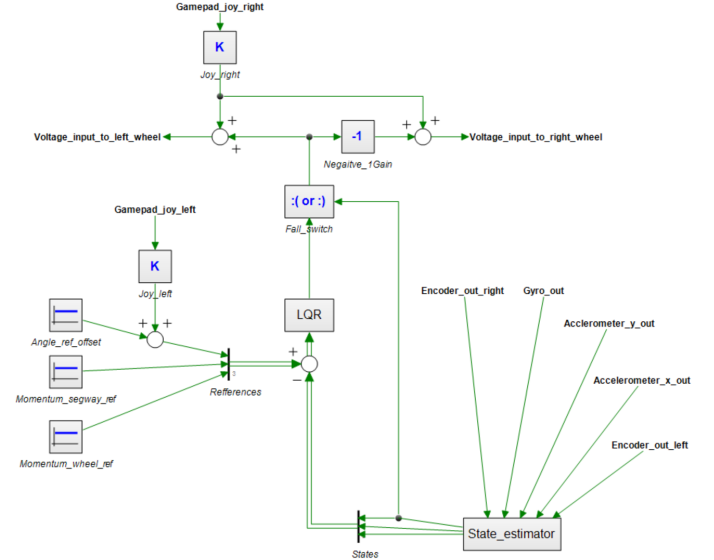


Fig. 10: Updated feedback implementation of the system with two gamepad joystick inputs. The input connected to the references is responsible for forwards and backwards movement, and that by the voltage inputs to the motors is responsible for turning

This setup also includes a fall switch which detects when the angle of the segway exceeds $1rad$, after which the segway most certainly cannot recover. Once this switch is triggered it simply switches off the LQR output until the angle is no longer exceeded, allowing for the segway to be re-lifted. This saves battery and will make the segway easier to test without the risk of fast-spinning wheels damaging the floor or someone's thumb.

III. METHODOLOGY

A. Model

Once the model has been constructed, its dynamic behaviour to an input can be used to verify its accuracy. To achieve this, an angle of 15° degrees is set as the initial condition in the integrator block. The expected result is an oscillation, which would simulate the segway falling over. In addition, a STL file of the frame and wheels has been uploaded to 20-sim to

³Single input single output

model the segway falling over in 3D. This can be seen in 11, and in the 20-sim file named "group07Segwayproject".

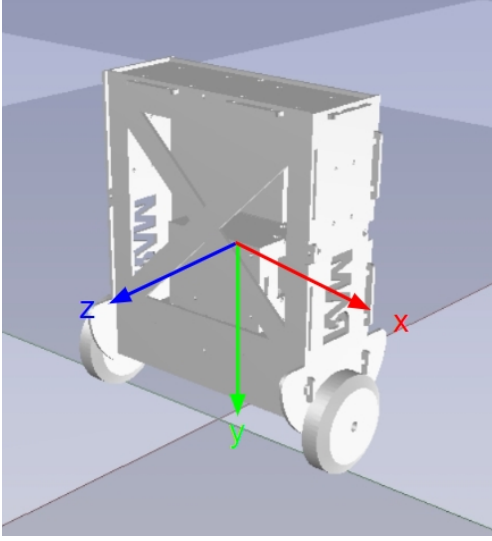


Fig. 11: 3D simulation of segway in 20 sim

The model is said to be valid if the period of oscillations is similar to the period that the segway experiences in real life. To validate this, the segway will be fixed by the motor axis and made to oscillate by displacing it with the same initial condition. The encoder's angle will then be recorded and compared with the simulation.

The model does not account for air resistance, therefore the period of the simulation is expected to be smaller than that of the encoder data. The oscillation is also expected to lose amplitude with time due to the mechanical friction and resistors modelled in the motors. However, the encoder data should experience this drop even faster due to air resistance, which is not accounted for in the simulation.

B. Characterization

There are a few segway parameters that need to be determined to accurately model the dynamic behaviour of the segway. As mentioned above, the model design and parameters of the two brushed MC33926PNB DC motors have been reported in [2] and will not be revisited in this paper. The remaining parameters that need to be determined are, the body's moment of inertia (I_b), the centre of mass along y-axis (y_{COM}), mass of segway body (m_b), mass of wheel (m_w), and radius of wheel (r_w).

1) *Center of Mass along y-axis (y_{COM}):* The centre of mass along y-axis (y_{COM}) is the distance from the bottom of the segway to its centre of mass, see Fig.1 as reference. the segway has irregular mass distribution due to the components inside, therefore this distance has to be determined experimentally. The method used for this application is to hang the segway by the wheel axis using a string, and attaching the other axis to a fixed body. In this orientation, the segway will act as a simply

supported beam, as illustrated in Fig.12, where position (x_1) is fixed.

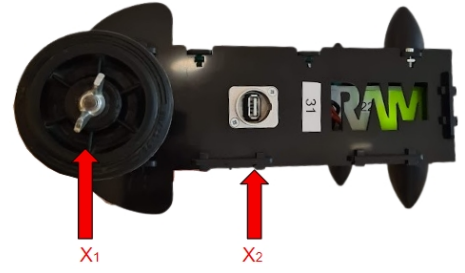


Fig. 12: Setup to determine the centre of mass along y-axis (wheels removed)

In this configuration, position (x_2) is moved until the segway is level with the ground. At that point, the distance between the two positions is the distance from the bottom of the segway to the centre of mass along the y-axis.

2) *Moment of inertia of segway body (I_b):* The segway is an irregular object with an uneven mass distribution. As a result, its moment of inertia about the motor shafts has to be determined experimentally. The physical pendulum theory [4] determines a body's moment of inertia around a pivot point, by displacing it from its equilibrium⁴ by an angle (θ). Consider Fig. 13

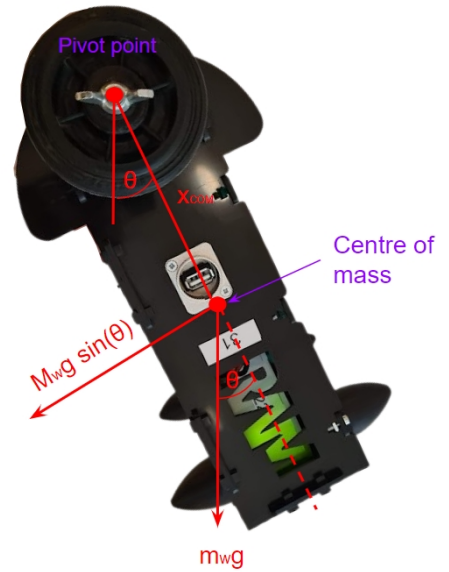


Fig. 13: Physical pendulum theory applied to segway, not to scale (wheels removed)

The dynamic behaviour of this system is given by equation 3:

⁴The equilibrium is the position in which the centre of mass is directly under the pivot point

$$I_b \frac{d^2\theta}{dt^2} = -y_{COM}(-m_b g) \sin(\theta) \quad (3)$$

Using small angle approximation ($\sin(\theta) = \theta$), the second-order linear ordinary differential equation can be solved giving the angular frequency (ω_0) shown in equation 4, where $\omega_0 = \frac{2\pi}{T}$.

$$\omega_0 = \sqrt{\frac{m_b \cdot g_{COM}}{I_b}} \quad (4)$$

Equation 4 can be rearranged to solve for moment of inertia, as seen in equation 5.

$$I_b = \frac{T^2 m_b \cdot g \cdot y_{COM}}{4\pi^2} \quad (5)$$

In practice, the moment of inertia is determined by mounting the segway at its axis and displacing it by an angle of 15° degrees to comply with small angle approximation. Once the segway is released, the period of oscillations T is then measured using the encoder of the motor. This method assumes that the pivot point is frictionless, however from the motor report [2], the mechanical friction of the shaft has been determined to be $1.02 \cdot 10^{-4} \text{N}$ which is assumed to be negligible.

3) *Other Parameters:* The mass of segway body (m_b) and mass of wheel (m_w) are determined using a precision scale. Lastly, the radius of wheel (r_w) is measured using a simple ruler.

C. State estimator

The validity of the state estimator depends on the sensor simulation and the properties of the sensor fusion implementation. Firstly, methods for validating the simulation will be described, and then those for the real estimator will also be defined. A working simulator is vital for easier initial tuning of the controller without the need for repeated uploading of the code to the segway.

1) *State sensor simulation:* The expected values for each of the two accelerometers, assuming a steady state, are: $g \cdot \cos(\theta)$ and $g \cdot \sin(\theta)$ for the Y and Z accelerometers respectively. A simple way to trial this such that the steady-state assumption holds is to reverse the direction of gravity in the model (only for segway movement, not sensors) and apply a constant voltage to generate a set offset. Once verified, applying a sinusoidal voltage input approximately equal to the natural frequency of the oscillation of the segway allows for the verification of the gyroscope simulation. This oscillation also introduces a translational velocity, the effect of which can be seen in the data, however, can be minimized given adjustments to the filtering cutoff frequency. Simply letting the segway oscillate in the downwards direction with correctly pointed gravity isn't as helpful due to the atan2 function's discontinuities and sensor simulators limiting the operating range of the state estimator to $\pm 90^\circ$.

Assuming correct values for the inertia of the segway, the state estimate of the angular momentum of the segway can be assumed to be precise due to a derivative not introducing

error to the same extent as an integral might with DC offset. Due to the angular momentum of the wheels being directly derived from the filtered angle, the validity can be assumed to propagate due to noise being verified as negligible in the Motor Report [2].

2) *Implemented state sensor:* Given that the wheel encoder can provide precise data for the movement of the segway, by fixing the wheels and measuring the initial angle, the change in angle can be used for a precise real-time angle reading of the tilt. With the wheels fixed, the segway can be tilted manually, and the accuracy of the filtered angle output can be verified. Using the same setup, the segway can be manually oscillated around the vertical equilibrium point to better validate the behaviour given small sharp oscillations, the type which are expected, and to be filtered by the complementary filter.

D. LQR

The LQR-based state-feedback controller aims to minimize the quadratic cost function 6 where x is the state variable and u the output [5]. The weighting matrix labelled Q is used to assign emphasis on certain state variables, and weighting factor.

$$J = \int_0^\infty [x^T(t)Qx(t) + u^T(t)Ru(t)] dt \quad (6)$$

There is an algebraic solution that leads to optimal Q and r_w values, however, due to the assumptions made in the model and measurement uncertainties in the parameters, this method will not yield valid results. Instead a more common approach of fixing the r_w value and adjusting weights of the Q matrix until specified closed loop performance specifications are met. If the system experiences sensor noise or large measurement error, a large r_w value is chosen [6].

To test Q matrix values, using the simulation, the system has been arranged into the feedback control form, shown in Fig. 9. The LQR gain values K are then calculated and inserted into the gain block labelled LQR. A simulation for an 15° degree initial angle is run, and the response is monitored. The control input voltage must remain below 24V to prevent the controller from overworking the motors by always providing the maximum voltage. Once, gain values that meet these requirements have been found in the simulated model, the controller is tested on the real segway.

IV. RESULTS

A. Model

Consider Fig. 14 below, showing the segway oscillating when subject to an 15° degree angle.

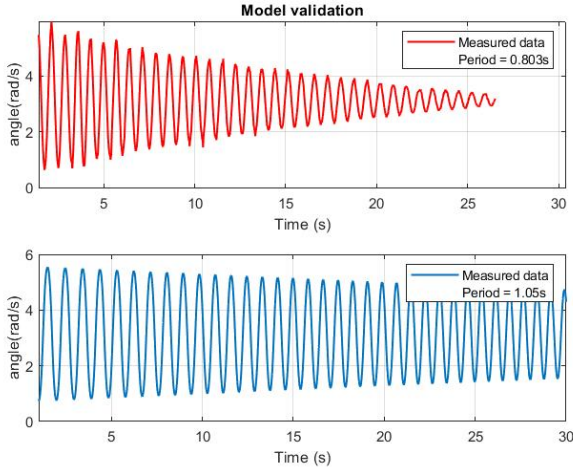


Fig. 14: Angle of segway in rad/s when subject to an 15° degree angle displacement

The period of oscillation for both the simulation and measured data is 1.05s and 0.803 respectively. The difference of 0.2s is assumed to be a result of air resistance. As predicted, the measured data experiences more damping than the simulation, which is also emphasised by the slower decrease in amplitude.

B. Characterization

The determined segway parameters can be seen in table I.

Values	Parameters
10.3	Distance to center of mass y_{COM} (cm)
0.0278	Moment of inertia of Body I_b (kgm ²)
4	Radius of wheel r_w (cm)
0.149	Mass of wheel m_w (kg)
2.20	Mass of segway m_b (kg)

TABLE I: Table with Segway parameters

The distance to the centre of mass was determined to be approximately 10.3 cm. The method used to determine this distance proved to be rather inaccurate, it can be said that it has an estimated error of ± 0.5 cm. To validate this value, the segway was able to successfully remain level along the y-axis when balanced at that point.

The mass of the wheel and the body were measured to be 0.149kg and 2.20kg respectively. Once these values had been collected, the segway was positioned as in the above described method and made to oscillate. The measured encoder data, allowed to determine the period of oscillations to be 0.803s. With this value and equation 5, the moment of inertia was calculated to be 0.0278 kgm². To validate this result, the moment of inertia can be calculated for a rod of the same length. Consider Fig. 15.

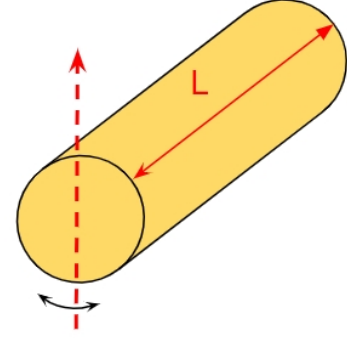


Fig. 15: Rod moment of inertia about the end

The moment of inertia is shown in equation, 7 [7].

$$I = \frac{1}{3}ML^2 \quad (7)$$

The moment of inertia for a rod of the same length is calculated to be 0.0466 kgm². Comparing the two values in depth is pointless due to differences in shape and mass distribution, however, both values are of the same order of magnitude. This provides some form of validity for the value determined experimentally.

Overall, the moment of inertia and distance to the centre of mass are not very accurate, since they exhibit some form of measurement error from the methods used to determine them. However, for the purpose of designing a controller, these errors can be neglected because the assumptions made in the model outweigh the impact of these measurement errors, making them insignificant. The controller will still require tuning after testing it on the real segway.

C. Model

Once a segway had been developed, a way to verify the accuracy of the simulated segway system versus the real one was to attempt to control the simulated system using real voltages and the same initial conditions. The plot of this validation can be seen in Fig. 16

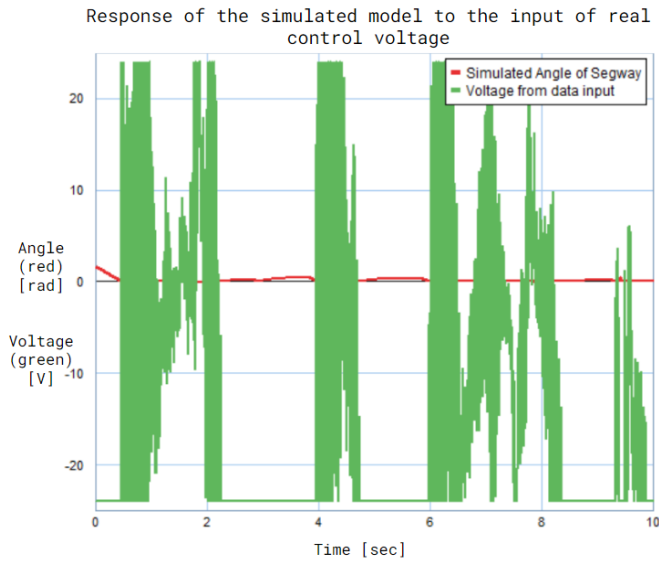


Fig. 16: The response of the simulated system given control voltage data collected from the real segway balancing

This system remains balanced remarkably well without the need for much tweaking of the initial simulated conditions as the segway was kept maximally upright when the start of the data collection. The position can be seen stabilising to the 0-position, followed by the position slowly drifting from the upright angle. This slow drifting in the angle is likely caused by an issue in the precision of the state estimator as also indicated by the fact that this drift only occurs in one direction.

D. Angle state Estimator

By fixing the wheels and measuring the steady-state values of the angle estimator as outlined in the methodology, the following validation plot can be generated to demonstrate the performance of the angle state estimator, see Fig. 17 18.

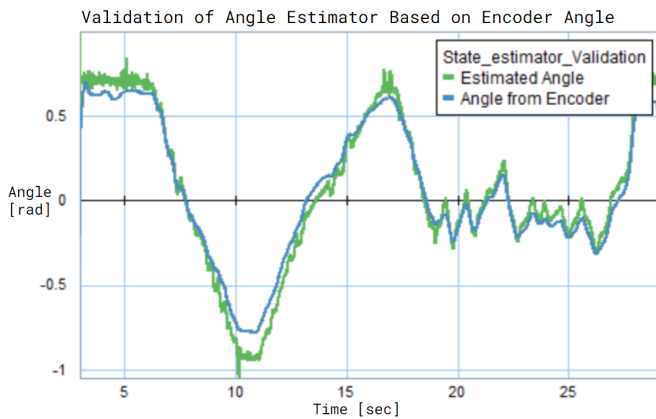


Fig. 17: Plot showing the angle output of the state estimator, versus the more reliable output of the wheel encoder

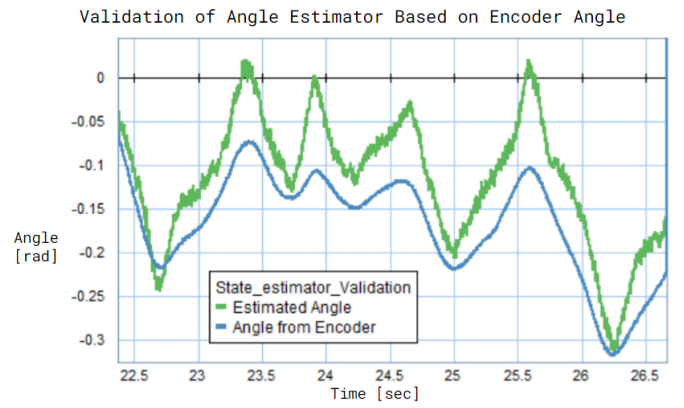


Fig. 18: Zoomed version of the equivalent plot above, showing maximum deviations of $-0.1\text{rad} \approx 6^\circ$

The state estimator is clearly able to take into account a majority of the noise approximated by the angle of the segway from the gyroscope and accelerometer inputs well. Although a deviation of approximately 0.1rad exists near the equilibrium, this could also be attributed to the method by which this data was collected not being ideal, namely, with the movement near equilibrium being faster than what would be present during operation. A more significant deviation from the values can be seen at points where the tilt exceeds 0.5 radians, however, this is beyond the likely operating range of the segway, and once this point has been reached, the segway is likely falling over, hence the need for precise angle estimation in this range is not present.

The deviation of 0.1rad consistently in one direction near the upright angle would also help explain the single-sided drift seen in the simulated angle as a response to real input data as seen in Fig. 16. From Fig. 17 it can be seen that this drift seems to increase the longer the data is collected, as no error near the equilibrium is present at the ≈ 8 second mark. This could be explained by not a sufficiently tuned complementary filter, however, from trials, increasing frequency any higher than the 8Hz used filters the gyroscope data too aggressively, filtered, leaving insufficient signal behind. One solution to this offset is to change the reference angle of the segway to take into account the difference, at the same time when tuning the controller. This could also be used to account for the slightly offset centre of mass in the Z-axis of the segway. Hence although a shortcoming of the estimator, can likely be easily worked around.

E. LQR

Since the r_w matrix defines the weight on the control that is input into the cost function 6, the larger the value, the less aggressive the controller. To determine the desired r_w value, a step response of the system with r_w equal to 0.01 and 1000 has been plotted. The Q matrix is kept as an identity matrix, consider Fig. 19.

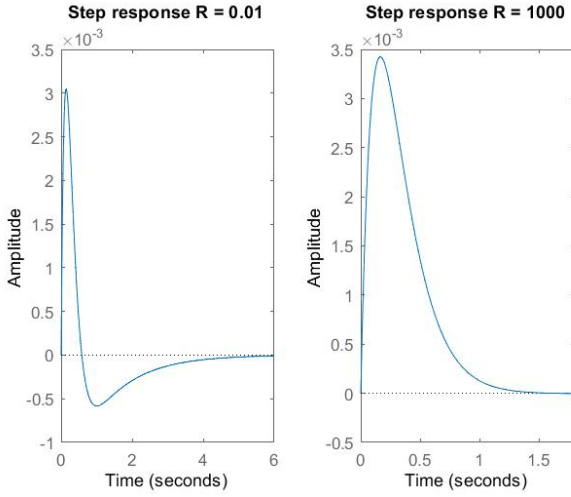


Fig. 19: Step response of controlled system with $r_w = 0.01$ and 1000. Q is set to an identity matrix

As shown in the figure, a controller with a r_w small value is too aggressive, which is seen by the undershoot. Therefore an r_w value of 1000 was chosen. In the Q matrix, a larger emphasis was made on the angle state, since that is also the output of the system. To test different weights, the LQR gains were applied to the 20-sim simulation to verify if they are able to control the system. The chosen values were able to stabilize a system for an 0.02 rad/s initial angle and stay within the specification of maximum 24V input. Take a look at Fig. 20.

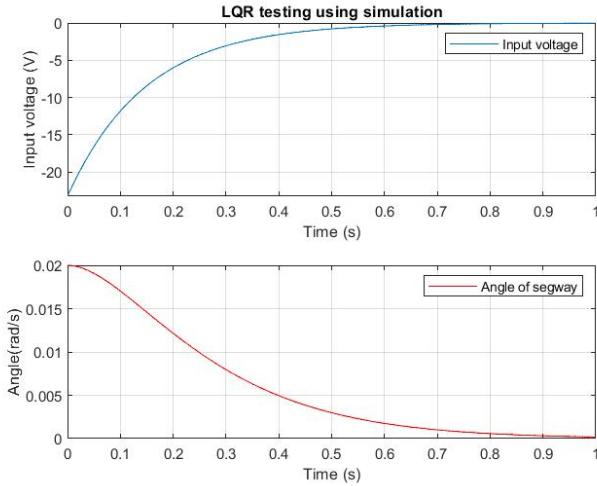


Fig. 20: LQR applied to simulation for in 0.02 rad/s initial angle.

As shown, the system stabilized the 0.02 rad/s angle offset and remained within 24V. The Q matrix for this controller is given by equation 8, where the states are in the following order; angle, momentum of wheel, momentum of segway.

$$Q = \begin{bmatrix} 10000 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 100 \end{bmatrix} \quad (8)$$

This weight placement resulted in the following LQR gain values K ;

$$K = [489.0 \quad 2161.1 \quad 0.031]$$

Applying this controller to the segway allowed caused it to stabilize and remain in an upright position, however, the segway would experience a significant amount of movement when stabilizing. Tuning the gain values by observing the real behaviour of the segway and adjusting accordingly resulted in a more stable controller. During the tuning process, an observation has been made that increasing the weight on the momentum of the segway, limits its translational movement in the Z-direction. This ensured that the segway would remain in a fixed radius for longer. The final K gain values are shown below.

$$K = [1163.27 \quad 6858.59 \quad 0.031]$$

As a way of validating the controller, the closed loop transfer function for the system was computed, consider equation 9, and its root locus in Fig. 21.

$$W(s) = \frac{0.05649s^2 + 2.408 \times 10^{-10}s - 1.242 \times 10^{-18}}{s^3 + 12.19s^2 + 37.46s + 1.078} \quad (9)$$

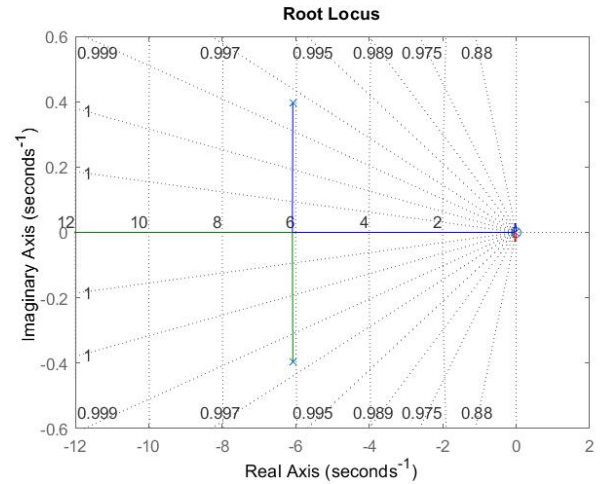


Fig. 21: Root locus of closed loop system $W(s)$, with poles at $-6.0790 + 0.3949i$, $-6.0790 - 0.3949i$, $-0.0290 + 0.0000i$

From the root locus, it can be seen that the closed-loop system is stable since there are no poles with a positive real part. This is also confirmed by the fact that segway is able to stand upright.

V. CONCLUSION

After all the aforementioned tests and investigations into the performance and tuning of the segway, a fairly successful segway was developed. First of all, its success in terms of the initially defined goals are as follows:

- **Stability while able to remain in a fixed radius**
The final segway was able to complete this challenge with no issue with a deviation of less than 10 cm over the course of the 30-second interval. This can be attributed to good controller design such that no oscillations were present so that the state estimator could focus on correcting very minor deviations in the position without the need for a separate position state to be taken into account in the controller.
- **Stability on an angled ramp or with external disturbances**
This challenge was only completed partially. Although the segway could withstand some minor disturbances such as driving over a small bump or a gentle push, it was not able to remain steady on an inclined plane.
- **Movement control with gamepad**
As expected, slightly different motor values resulted in fairly uneven turning when going forward, yet the segway was overall easy to manoeuvre and remained stable in doing so. Given that during movement, especially when changing speeds, the voltage sometimes clipped to the 24V margin, the use of the reference to control the speed as opposed to wheel voltage enabled the system to remain much more stable.

In terms of more precise validated values, the period of oscillation of the real segway was measured to as 0.803 seconds, whereas that of the simulation was found to be 1.05 seconds. With significantly more damping being present for the real data.

VI. DISCUSSION

Starting from the initially defined goals, the main challenge which was not fulfilled was the ability of the segway to remain steady on an inclined plane. During the tests on this plane, the segway would remain stable for only a few seconds before veering off and dismounting the platform. Due to a simple offset in the reference angle being used to account for both the difference in the reading of the angle by the state estimator as well as the offset centre of mass, the reading for the angle of the segway when inclined could be affected. Due to the offset, the centre of mass applies a different static torque on the body when on an inclined plane versus a flat surface, resulting in the offset in the reference is no longer relevant on the plane. The ideal way of approaching this issue would be to recalibrate the state estimator and conduct new measurements on noise frequencies and levels.

During the design of the controller, the R matrix is set to a large value. The thought process behind this decision was that it would limit the overshoot and prevent over-controlling the segway. In the future, it would be interesting to see whether increasing the R matrix would help with the stability on the angled ramp and with external disturbances. Furthermore, a

larger emphasis on staying below 24V should have been made, because as shown in Fig.16, the segway still maxes out at 24V sometimes, which is not ideal due to additional wear on the motors as well as noise and a risk of the motors slowly coming loose of their mounts due to excessive torques and vibrations.

A very important fact to consider and evaluate the success of this project is the similarity between the model and the real segway. As already stated, there was a significant difference in the period measured when verifying the moment of inertia of the segway, this is overruled by the extremely good results of Fig. 16 when real control voltage data was applied to the simulated model. Even if a difference in inertia was present, according to this test it had a negligible impact on the control of the system. This finding also allows the conclusion that the assumptions made (no inductance, no slip, no air resistance) had little impact on the dynamic behaviour of the system when it comes to controllability and controller design. This sort of consistency between reality and simulation was very surprising and therefore allows for the attention to potential sources of issues to be directed to more likely sources.

Alongside the offset of the state sensor, its performance was likely the primary limiting factor in the performance of the segway. In practice, the poor performance became especially apparent when driving the segway at higher speeds and when attempting to slow the segway down quickly by applying a signal to move in the opposite direction while still moving forward. This could be brought down to poor optimisation of the filtering within the sensor fusion, and more likely that of the accelerometers. Increasing the cutoff frequency of the complementary filter would have potentially improved the performance of the atan2 angle approximation at some cost of delay, but would have reduced the output of the gyroscope even further which would have made it have close to no impact on the overall angle estimate. If this project was to be continued, the possibility of having two different cutoff frequencies alongside weighted sums of the two angle sensors might allow for some middle ground of all the advantages of each setup to be found, by perhaps increasing the cutoff for the filter of the accelerometer, but decreasing that of the gyroscope.

REFERENCES

- [1] W. Yannik, K. Dimitris, and K. Gijs, *Lab Manual Segway Project*, 2022-2023.
- [2] E. Grava and S. Letkiewicz, "Motor Report," Tech. Rep., Dec. 2022.
- [3] S. Chaudhary and A. Kumar, *Control of Twin Rotor MIMO System using PID and LQR Controller*. Journal of Aircraft and Spacecraft Technology, Aug. 2019.
- [4] "Physical Pendulum: Finding Moment of Inertia | PocketLab." [Online]. Available: <https://archive.thepocketlab.com/educators/lesson/physical-pendulum-finding-moment-inertia>
- [5] S. Das, I. Pan, K. Halder, S. Das, and A. Gupta, "Optimum weight selection based lqr formulation for the design of fractional order pid controllers to handle a class of fractional order systems," in *2013 International Conference on Computer Communication and Informatics*, 2013, pp. 1–6.
- [6] A. Noshadi, "How to determine the values of the control matrices q and r for the lqr strategy when numerically simulating the semi-active tldc." 10 2015.

- [7] P. A. Tipler and G. Mosca, *Physics for scientists and engineers: with modern physics*, sixth edition, [extended version, ch. 1-41, r] ed. New York, NY Basingstoke: W.H. Freeman and Company, 2008.